Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-C Ischemic Heart Disease

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-03-C01-BD1

Brain dump 1 · foundational retrieval: Myocardial oxygen supply and demand from memory

A competent response contains

· Resting coronary flow is about 225 mL/min, about 4 to 5% of cardiac output, and can rise about three to four fold in heavy exercise.

· The myocardium extracts about 70% or more of the oxygen in coronary blood at rest, so extra demand can be met only by raising flow.

· Flow reserve comes from dilation of small resistance arterioles in response to local metabolic signals: chiefly adenosine from ATP breakdown, with low oxygen tension, carbon dioxide, hydrogen ions, and potassium.

· Demand is set by heart rate, contractility, and ventricular wall stress.

· By the law of Laplace, wall stress is proportional to pressure times radius divided by wall thickness, so high afterload or a large chamber raises oxygen use.

· The left ventricle is perfused mainly in diastole, and the driving pressure is approximately aortic diastolic pressure minus left ventricular end-diastolic pressure.

· The subendocardium is ischemic first because systolic compression of intramural vessels is greatest there and penetrating branches reach it last.

· Subendocardial ischemia therefore appears whenever perfusion pressure falls, diastole shortens, or ventricular diastolic pressure rises.

Distinctions that matter

· Ischemia is inadequate flow, which removes oxygen and leaves metabolic waste behind; hypoxemia lowers oxygen content while flow continues.

· Coronary flow reserve is a property of the arterioles, while a fixed stenosis is a property of a large epicardial artery; the arterioles can hide the stenosis at rest.

Common misconceptions

· Expecting the heart to meet higher demand by extracting more oxygen. Extraction is already near maximal at rest, so flow has to rise.

· Treating ischemia as a disease of one artery. It is a mismatch between supply and demand, which is why aortic stenosis and severe anemia cause angina with normal coronary arteries.

Where it is taught: Section C01, first three paragraphs and the oxygen balance table of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C01-BD2

Brain dump 2 · integration and application: Four patients with chest pain and clean coronary arteries

A competent response contains

· Patient 1, aortic stenosis with hypertrophy: both sides. High systolic pressure and hypertrophy raise wall stress and oxygen use per beat, while a long ejection time and a high left ventricular end-diastolic pressure cut the subendocardial driving pressure, and hypertrophy adds muscle with fewer capillaries per myocyte.

· Patient 2, hemoglobin 6.5 g/dL: supply side through oxygen content. Each mL of coronary blood carries less oxygen, so the same flow delivers less oxygen; compensatory tachycardia adds a demand problem and shortens diastole.

· Patient 3, hyperthyroidism at 136/min: demand side mainly. Heart rate and contractility raise oxygen use per minute, and tachycardia also shortens diastole, so perfusion time falls.

· Patient 4, cocaine: supply side through coronary resistance. Vasospasm of an epicardial artery, with added sympathetic rise in rate and blood pressure on the demand side.

· Patients 1 to 3 develop subendocardial ischemia, because that layer is compressed in systole and perfused last, so it suffers first whenever supply falls short.

· Patient 4 can develop ischemia through the whole wall thickness, because spasm of an epicardial artery shuts off flow to the full thickness of its territory rather than trimming the reserve.

Distinctions that matter

· A demand problem spends the reserve from above; a content problem lowers the oxygen carried in a normal flow; a resistance problem blocks flow itself.

· Layer follows mechanism: reduced perfusion pressure or shortened diastole starves the subendocardium, while epicardial occlusion or spasm takes the whole wall.

Common misconceptions

· Concluding that normal coronary arteries rule out myocardial ischemia. Aortic stenosis, severe anemia, hypoxemia, carbon monoxide poisoning, tachycardia, and spasm all cause it with patent arteries.

· Assigning each patient to one row of the oxygen balance table. Most triggers act on several rows at once, and tachycardia almost always acts on both supply and demand.

Where it is taught: Section C01, oxygen balance table and its caption of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C01-PR01

Prediction exercise: Predict what happens when demand rises on a fixed stenosis

A competent response contains

· Oxygen content per mL of blood falls, because hemoglobin is 7.2 g/dL and hemoglobin carries almost all the oxygen.

· Perfusion time falls, because a rate of 134/min shortens diastole, and the left ventricle is perfused mainly in diastole.

· The gradient for subendocardial flow falls, because the driving pressure is aortic diastolic pressure minus left ventricular end-diastolic pressure and the blood pressure is 86/50 mm Hg.

· Oxygen used per minute rises, because fever, tachycardia, and sympathetic drive raise heart rate and contractility.

· Arteriolar reserve distal to the stenosis is already spent, because the arterioles had dilated to keep resting flow normal across a 75% narrowing, so no further dilation is available.

· The subendocardium becomes ischemic first, and the leads facing that territory show ST depression, because the injury current points toward the cavity and away from the surface electrodes.

Distinctions that matter

· Fever and tachycardia act on both sides of the balance at once: more oxygen used per minute and less time to deliver it.

· Anemia lowers oxygen content without changing resistance, so flow can look adequate while delivery is not.

Common misconceptions

· Expecting arteriolar dilation to rescue this situation. The reserve was already used to keep resting flow normal past the stenosis.

· Looking for ST elevation here. Subendocardial ischemia depresses the ST segment, and that depression does not localize the artery.

Where it is taught: Section C01, oxygen balance table; Section C02, paragraphs on stable angina and ST segment movement of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C02-BD1

Brain dump 1 · foundational retrieval: Stable angina and the moving ST segment

A competent response contains

· Angina pectoris is episodic chest pressure or heaviness from transient ischemia that is not severe or long enough to kill myocytes.

· Ischemic myocytes release adenosine, bradykinin, hydrogen ions, and potassium, which stimulate visceral afferents travelling with the cardiac sympathetic nerves into roughly T1 to T5.

· The same dorsal horn neurons receive somatic input from the chest wall and inner arm, so the pain is referred to the chest, left arm, neck, and jaw.

· Stable angina comes from a fixed stenosis of about 70% or more, and about 90% or more can limit flow at rest.

· By the Poiseuille relationship resistance varies inversely with the fourth power of the radius, so halving the radius raises segment resistance 16 fold.

· At rest the downstream arterioles dilate enough to keep flow normal, which spends the reserve, so flow cannot rise with exertion and the subendocardium becomes ischemic; pain is predictable, brief, and eased by rest.

· Nitroglycerin works because nitric oxide raises cGMP and dilates veins more than arteries, lowering ventricular volume, wall stress, and demand.

· The ST segment is normally isoelectric because all healthy ventricular cells sit at the same plateau voltage, so no gradient exists to record.

· Ischemic cells lose potassium, depolarize partially, and have shorter lower plateaus, so an injury current flows between ischemic and normal muscle.

· Subendocardial ischemia points the ST vector toward the cavity and away from surface electrodes, giving ST depression that does not localize the artery; ischemia through the whole wall thickness points it toward the epicardium, giving ST elevation in the facing leads, which does localize the artery.

Distinctions that matter

· Flow reserve is spent at rest in stable angina, so resting flow is normal while exertional flow is not; this is why the history is exertional rather than constant.

· ST depression reports the layer, not the artery, while ST elevation reports both.

Common misconceptions

· Explaining nitroglycerin by coronary dilation alone. Its main action here is venodilation that lowers ventricular volume, wall stress, and demand.

· Reading an isoelectric ST segment as electrical silence. The whole ventricle is depolarized at the plateau, with no voltage gradient to record.

Where it is taught: Section C02, opening paragraph, stable angina paragraph, and the paragraph on why the ST segment moves of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C02-BD2

Brain dump 2 · integration and application: Sort three episodes of chest pain by mechanism

A competent response contains

· Patient 1: stable angina from a fixed stenosis of about 70% or more; subendocardial ischemia; ST depression during pain; troponin normal. The reserve is spent at rest, so flow cannot rise with exertion and the inner layer starves first.

· Patient 2: unstable angina, the mildest acute coronary syndrome, from a fissured plaque with a nonocclusive platelet thrombus, sometimes with spasm from thromboxane A2 and serotonin; usually subendocardial ischemia; ST depression or T inversion or a normal tracing; troponin stays below the upper reference limit.

· Patient 3: Prinzmetal (variant) angina from intense spasm of an epicardial artery, with or without plaque, linked to smooth muscle hyperreactivity and reduced endothelial nitric oxide; ischemia through the full wall thickness, so the ST segment rises transiently; troponin normal unless infarction occurs.

· A troponin rise above the assay-specific 99th percentile upper reference limit would make patient 2 a non-ST-elevation myocardial infarction.

· Prinzmetal attacks come at rest, often at night, in younger patients whose main risk factor is smoking, and can be provoked by cocaine, amphetamines, ergot alkaloids, triptans, and hyperventilation.

· Microvascular angina causes exertional ischemia with normal epicardial arteries and belongs in the same differential.

Distinctions that matter

· Unstable angina and non-ST-elevation myocardial infarction share the lesion and the tracing; serial troponin is what separates them.

· Rest pain with ST elevation that comes and goes is spasm, while rest pain with ST depression is an active nonocclusive plaque.

Common misconceptions

· Taking transient ST elevation as proof of infarction. In Prinzmetal angina it resolves with the spasm and troponin stays normal.

· Calling unstable angina a mild problem because troponin is normal. The thrombus can grow and occlude the artery.

Where it is taught: Section C02, unstable, Prinzmetal, and microvascular paragraph, the comparison table, and the stop and predict box of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C02-RT01

Retrieval exercise: Rebuild the ischemic cascade

A competent response contains

· Step 1: demand exceeds supply, or supply suddenly falls.

· Step 2: anaerobic glycolysis begins and lactate, hydrogen ions, and adenosine accumulate.

· Step 3: relaxation slows, because calcium reuptake needs ATP.

· Step 4: regional contraction weakens.

· Step 5: the ST segment changes from injury currents.

· Step 6: chest pain follows from stimulation of visceral afferents.

· The pump that needs ATP at step 3 is the sarcoplasmic reticulum calcium ATPase, which pulls calcium back out of the cytosol so the muscle can relax.

· At step 5 ischemic cells have lost potassium and are partly depolarized with shorter lower plateaus, so a current flows between ischemic and normal myocardium and displaces the ST segment.

· Because pain is last in the order, ischemia can be silent, especially when diabetic autonomic neuropathy has damaged the afferent fibers.

· Rest restores the balance between supply and demand and the whole cascade reverses without necrosis.

Distinctions that matter

· Relaxation fails before contraction does, because taking calcium back up costs ATP while letting the muscle relax is not passive.

· Electrical change precedes pain, so a normal tracing taken between episodes does not rule the condition out.

Common misconceptions

· Putting pain early in the cascade. It is the last event, which is what makes silent ischemia possible.

· Assuming any episode of ischemia kills muscle. In angina the episode ends before myocytes die.

Where it is taught: Section C02, the ischemic cascade flow diagram and its caption of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C02-CP01

Comparison table: Stable, unstable, and Prinzmetal angina

A competent response contains

· Lesion: stable angina has a fixed stenosis of about 70% or more; unstable angina has a disrupted plaque with a nonocclusive thrombus; Prinzmetal angina is epicardial vasospasm.

· Pattern: stable is exertional, predictable, and relieved by rest; unstable is rest pain, new pain, or crescendo pain; Prinzmetal is spontaneous rest pain, often at night.

· Layer ischemic: subendocardial in stable, usually subendocardial in unstable, and the full wall thickness in Prinzmetal.

· ECG during pain: ST depression in stable; ST depression, T inversion, or a normal tracing in unstable; transient ST elevation in Prinzmetal.

· Troponin: normal in stable; normal in unstable, which is what separates it from non-ST-elevation myocardial infarction; normal in Prinzmetal unless infarction occurs.

· Patient and provoking factors: stable angina in a patient with established coronary disease provoked by exertion, cold, or emotion; unstable angina from plaque disruption with thromboxane A2 and serotonin driving added spasm; Prinzmetal in younger patients whose main risk factor is smoking, provoked by cocaine, amphetamines, ergot alkaloids, triptans, and hyperventilation.

· Main risk: gradual progression in stable angina, infarction in unstable angina, and arrhythmia or infarction in Prinzmetal angina.

Distinctions that matter

· The ischemic layer decides whether the ST segment falls or rises, which is why Prinzmetal angina elevates it and the other two depress it.

· Stable and unstable differ by the state of the plaque rather than by the degree of narrowing; many culprit plaques are only mildly to moderately stenotic.

Common misconceptions

· Grading these three by severity of stenosis. Plaque stability, not size, decides which syndrome a patient has.

· Expecting Prinzmetal angina only in patients with atherosclerosis. Spasm occurs with or without plaque.

Where it is taught: Section C02, comparison table of stable, unstable, and Prinzmetal angina of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C03-BD1

Brain dump 1 · foundational retrieval: From a ruptured cap to a dead myocyte

A competent response contains

· Sequence: the cap ruptures and exposes collagen and a lipid core rich in tissue factor; von Willebrand factor binds platelet glycoprotein Ib and platelets adhere; ADP and thromboxane A2 recruit more platelets and glycoprotein IIb/IIIa binds fibrinogen; tissue factor drives thrombin and fibrin and the thrombus occludes the artery.

· Contraction stops within about 1 to 2 minutes, irreversible injury begins after about 20 to 40 minutes starting in the subendocardium, and necrosis spreads toward the epicardium over about 3 to 6 hours.

· Inside the myocyte: oxidative phosphorylation stops and anaerobic glycolysis yields little ATP while generating lactate and hydrogen ions.

· Failure of the sodium-potassium ATPase lets sodium and water in so the cell swells, potassium leaks out, and calcium accumulates as ATP-dependent pumps fail.

· Calcium floods in and activates phospholipases and proteases.

· The two changes marking irreversible injury are sarcolemmal disruption and amorphous densities in mitochondria.

· Troponin, the MB isoenzyme of creatine kinase, and myoglobin leak into blood.

· If flow returns within roughly 20 minutes the injury is reversible, although the muscle may stay stunned for hours to days.

· Rupture clusters in the early morning and after intense exertion or emotional stress, when an adrenergic surge raises blood pressure, heart rate, and shear stress.

· Most infarctions follow rupture of a thin inflamed cap; fewer follow superficial erosion over an intact plaque, and many culprit plaques were only mildly to moderately stenotic, so infarction is often the first sign of coronary disease.

Distinctions that matter

· Loss of contraction at 1 to 2 minutes is a functional failure in living cells, while irreversible injury at 20 to 40 minutes is cell death; the gap between them is the window for salvage.

· Plaque rupture exposes tissue factor and collagen, so both platelet adhesion and the coagulation cascade start at the same lesion.

Common misconceptions

· Assuming the most stenotic plaque is the one that infarcts. Cap thickness, lipid core size, and inflammation decide rupture, not the degree of narrowing.

· Treating the immediate wall motion abnormality as proof of necrosis. Contraction fails long before the cell dies.

Where it is taught: Section C03, opening paragraph, the plaque rupture flow diagram, and the paragraphs on the ischemic myocyte of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C03-BD2

Brain dump 2 · integration and application: Two infarcts and two reperfused hearts

A competent response contains

· Eight hours of complete occlusion gives a transmural infarct that spans the full or nearly full wall in one artery's territory, with ST elevation, later pathologic Q waves, a troponin rise, and the electrical label of ST-elevation myocardial infarction.

· Thirty minutes of partial occlusion that then opens gives a subendocardial infarct limited to the inner third to half of the wall, with ST depression or T inversion, usually no Q waves, a troponin rise, and the label of non-ST-elevation myocardial infarction.

· The transmural infarct can cause pericarditis and free wall rupture because necrosis reaches the epicardium; the subendocardial infarct rarely ruptures because the outer wall stays intact.

· The electrical and pathologic labels are strongly associated but not identical.

· Reperfused infarcts are often hemorrhagic on gross examination.

· Irreversibly injured myocytes take up a surge of calcium and hypercontract, producing contraction band necrosis with dense eosinophilic cross bands.

· Reactive oxygen species and calcium opening the mitochondrial permeability transition pore kill some cells that might have survived, which is reperfusion injury, so early reperfusion limits infarct size while changing the appearance of the infarct.

· In shock or profound hypoxemia a circumferential subendocardial infarct ignores arterial territories, because the whole inner layer is underperfused rather than one artery being blocked.

Distinctions that matter

· Transmural against subendocardial is a statement about the tissue; ST elevation against no ST elevation is a statement about the tracing, and the two pair up without being the same thing.

· Infarct size is limited by reperfusion, while the appearance of the infarct is altered by it; hemorrhage and contraction bands are features of the reperfused infarct.

Common misconceptions

· Equating every ST-elevation infarct with a transmural one at autopsy. The pairing is an association.

· Treating reperfusion as purely protective. It salvages muscle overall and also kills some borderline cells through oxygen radicals and calcium overload.

Where it is taught: Section C03, paragraphs on transmural and subendocardial infarction and on reperfusion of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C03-RT01

Retrieval exercise: Rebuild the path from plaque rupture to transmural infarction

A competent response contains

· Step 1: the cap ruptures, exposing collagen and a lipid core rich in tissue factor.

· Step 2: von Willebrand factor binds platelet glycoprotein Ib and platelets adhere to the exposed surface.

· Step 3: ADP and thromboxane A2 recruit further platelets, and glycoprotein IIb/IIIa binds fibrinogen to cross-link them.

· Step 4: tissue factor from the lipid core drives thrombin and fibrin generation and the thrombus occludes the artery.

· Step 5: contraction stops within about 1 to 2 minutes.

· Step 6: irreversible injury begins after about 20 to 40 minutes, in the subendocardium first.

· Step 7: necrosis spreads toward the epicardium over about 3 to 6 hours.

· Shading: at 30 minutes only the inner third to half is dying; at 2 hours the necrosis has moved outward into the middle of the wall; by about 6 hours it reaches the epicardium in the territory supplied.

· Collateral flow and reperfusion can stop the wavefront before it becomes transmural.

Distinctions that matter

· Platelet adhesion needs von Willebrand factor and glycoprotein Ib, while platelet aggregation needs glycoprotein IIb/IIIa and fibrinogen; the drawing should separate the two sites.

· The wavefront moves from endocardium outward, so the outer layers have the longest window for salvage.

Common misconceptions

· Drawing the necrosis beginning in the middle of the wall or spreading inward. It starts in the subendocardium and moves out.

· Leaving the lipid core out of the clotting story. Tissue factor in the core is what drives thrombin generation.

Where it is taught: Section C03, flow diagram from plaque rupture to transmural infarction and its caption of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C04-BD1

Brain dump 1 · foundational retrieval: Build the coronary territory table from memory

A competent response contains

· Anterior: left anterior descending artery, proximal if extensive; ST elevation in V1 to V4 and also V5, V6, I, and aVL if extensive; largest infarcts, with shock, bundle branch block, apical thrombus, and aneurysm.

· Lateral: left circumflex artery or an LAD diagonal branch; ST elevation in I, aVL, V5, and V6; reciprocal inferior ST depression.

· Inferior: right coronary artery in most hearts, circumflex if left dominant; ST elevation in II, III, and aVF; bradycardia, atrioventricular block, and papillary muscle rupture.

· Right ventricular: proximal right coronary artery; ST elevation in V4R with inferior changes; hypotension, clear lungs, and raised neck veins.

· Posterior: right coronary or circumflex; no elevation in the standard leads, with a mirror image of ST depression and tall R waves in V1 to V3; occurs with inferior or lateral infarction.

· The LAD supplies the anterior left ventricular wall, the apex, and the anterior two thirds of the septum with much of the bundle branches; the circumflex supplies the lateral wall; the right coronary artery supplies the right ventricle, the inferior left ventricular wall, and usually the sinoatrial and atrioventricular nodes.

· The posterior descending artery arises from the right coronary artery in about 80% of hearts, which is right dominance, and otherwise from the circumflex.

· The LAD is occluded most often, about 40 to 50% of cases, then the right coronary and the circumflex.

· Reciprocal ST depression supports infarction over pericarditis, which elevates the ST segment diffusely without reciprocal change.

· A pathologic Q wave appears because dead myocardium is electrically silent, so an electrode over it records the opposite wall depolarizing away; it is commonly at least 40 ms wide or more than about one quarter of the R wave height.

Distinctions that matter

· Lead groups localize the artery only when the ST segment is elevated; depression from subendocardial ischemia does not localize.

· Posterior infarction is read as a mirror image in the anterior leads rather than as elevation anywhere on the standard tracing.

Common misconceptions

· Assigning the inferior wall to the right coronary artery in every patient. In a left dominant heart the circumflex supplies it.

· Reading ST elevation in many lead groups at once as a large infarct without checking for reciprocal change. Diffuse elevation without reciprocal depression points to pericarditis.

Where it is taught: Section C04, first paragraph, the territory table, and the paragraph on reciprocal change and Q waves of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C04-BD2

Brain dump 2 · integration and application: Read the biomarkers across four days

A competent response contains

· Troponin I or T first rises at about 3 to 4 hours, earlier with high-sensitivity assays, peaks at about 24 to 48 hours, and returns to normal at about 7 to 10 days, with troponin T up to about 14 days.

· The MB isoenzyme of creatine kinase first rises at about 3 to 6 hours, peaks at about 24 hours, and returns to normal at about 48 to 72 hours.

· Myoglobin first rises at about 1 to 3 hours, peaks at about 6 to 12 hours, and returns to normal at about 24 hours, so it is earliest but nonspecific.

· Troponin persists because a small cytosolic pool leaks early and a large myofilament-bound pool is released slowly; the MB isoenzyme is cytosolic and clears quickly.

· On day 4 troponin is still high from the first infarct, so a single value cannot show new necrosis, while the MB isoenzyme had normalized by about 48 to 72 hours, so a new rise in it points to reinfarction.

· If the MB isoenzyme is normal and the pain is pleuritic and positional, fibrinous pericarditis is the more likely explanation.

· Troponin also rises in myocarditis, pulmonary embolism, sepsis, heart failure, tachyarrhythmia, cardiac contusion, and chronic kidney disease.

· Infarction requires a rise or fall of troponin with at least one value above the assay-specific 99th percentile, plus evidence of ischemia.

Distinctions that matter

· Specificity and usefulness for reinfarction run in opposite directions: troponin is the most specific marker but too persistent, while the MB isoenzyme is less specific but clears fast enough to show a second event.

· A high troponin reports myocyte necrosis of any cause; infarction is necrosis from ischemia, which is why the definition asks for a rise and fall with evidence of ischemia.

Common misconceptions

· Reading a single high troponin on day 4 as reinfarction. It can simply be the tail of the first event.

· Taking any troponin rise as an infarction. Pulmonary embolism, sepsis, myocarditis, and chronic kidney disease all raise it.

Where it is taught: Section C04, biomarker paragraph, the biomarker table, and the worked example on recurrent chest pain on day 4 of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C04-Q01

Board-style question: Hypotension with clear lungs after an inferior infarction

Correct answer: C. Right ventricular infarction limiting delivery of blood to the left heart

Why it is right: A proximal right coronary occlusion infarcts the right ventricle as well as the inferior left ventricular wall. The right ventricle cannot pump its volume through the lungs to the left heart, so left ventricular filling and cardiac output fall while the lungs stay clear and systemic venous pressure rises, and ST elevation in V4R is the finding that identifies it.

A is wrong: Papillary muscle rupture with acute mitral regurgitation floods the lungs, so the examination would show widespread crackles rather than clear lung fields, and it appears around days 3 to 7 rather than within 2 hours of the onset of pain.

B is wrong: Blood in the pericardial sac follows free wall rupture of a transmural infarct, usually around days 3 to 7, and would present with pulseless electrical activity rather than a measurable blood pressure 2 hours into the first episode of pain.

D is wrong: Loss of more than 40% of left ventricular muscle causes cardiogenic shock with pulmonary congestion and crackles, and that degree of loss follows a large anterior infarct rather than an inferior one.

A competent response contains

· ST elevation in II, III, and aVF with reciprocal depression in I and aVL is an inferior infarct, usually from right coronary artery occlusion.

· Elevation in V4R adds right ventricular infarction, which occurs with proximal right coronary occlusion.

· A failing right ventricle cannot deliver blood to the left heart, so filling pressures rise behind it while the pulmonary circulation stays dry: hypotension, raised jugular venous pressure, and clear lungs.

· The bradycardia fits right coronary supply of the sinoatrial and atrioventricular nodes together with vagal reflexes from the inferior and posterior wall.

Distinctions that matter

· Hypotension with clear lungs points to the right ventricle, while hypotension with crackles points to loss of left ventricular muscle.

Common misconceptions

· Reading raised neck veins after an infarction as tamponade in every case. Right ventricular infarction raises them without any pericardial blood.

Where it is taught: Section C04, territory table; Section C06, paragraph on pump failure and right ventricular infarction of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C05-BD1

Brain dump 1 · foundational retrieval: Build the healing timeline from memory

A competent response contains

· 0 to 30 minutes: no gross change; no light microscopic change, with reversible changes visible by electron microscopy; arrhythmia.

· 30 minutes to 4 hours: no gross change, pale on tetrazolium stain after about 2 to 3 hours; usually no light microscopic change, with wavy fibers at the border; ventricular fibrillation and sudden death.

· 4 to 24 hours: occasional dark mottling; early coagulative necrosis, edema, hemorrhage, contraction bands, early neutrophils; arrhythmia, left ventricular failure, cardiogenic shock.

· 1 to 3 days: mottling with a yellow-tan center; complete coagulative necrosis, loss of nuclei, brisk neutrophils; fibrinous pericarditis.

· 3 to 7 days: hyperemic border with central softening; disintegrating myocytes with macrophages at the border; free wall, septal, or papillary muscle rupture.

· 7 to 10 days: softest and most yellow with red-tan margins; phagocytosis and early granulation tissue at the margins; rupture risk persists.

· 10 to 14 days: red-gray depressed borders; established granulation tissue with new vessels and collagen; rupture less likely.

· 2 to 8 weeks: gray-white scar forming from the border inward; increasing collagen and decreasing cellularity; aneurysm, mural thrombus, Dressler syndrome.

· Over 2 months: firm white scar; dense collagen; heart failure and scar-related ventricular tachycardia.

· Dead myocytes keep their outlines for several days because ischemia and acidosis denature structural proteins and lytic enzymes alike, so autolysis is delayed; this is coagulative necrosis.

· Adult myocytes have very little capacity to divide, so dead muscle is replaced by scar rather than new muscle.

· Healing advances from the viable margins inward, so a large infarct heals more slowly and its center lags behind its edges.

· Wavy fibers appear at the border within hours because contracting living fibers stretch the noncontracting dead ones.

· Triphenyltetrazolium chloride stains viable muscle brick red through dehydrogenase activity, so an infarct appears as a pale unstained zone after about 2 to 3 hours, before light microscopy shows necrosis.

· Neutrophils arrive near the end of day 1, drawn by chemokines and complement fragments; contraction bands appear at the margins in the first day and throughout reperfused infarcts.

Distinctions that matter

· Neutrophils peak in the first 3 days and macrophages take over from about day 3 to day 7; the cell type present dates the infarct more reliably than the gross appearance.

· Loss of dehydrogenase activity precedes visible necrosis, which is why tetrazolium staining detects an infarct that light microscopy would call normal.

Common misconceptions

· Expecting dead myocytes to disappear quickly. In coagulative necrosis their outlines persist for days because the lytic enzymes were denatured too.

· Looking for granulation tissue in the first week. It appears at the margins around days 7 to 10 and is established by about days 10 to 14.

Where it is taught: Section C05, opening paragraphs and the morphologic timeline table of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C05-BD2

Brain dump 2 · integration and application: Date four infarcts from what the tissue shows

A competent response contains

· Heart 1 is about 3 to 7 days old: macrophage clearance with central softening and a hyperemic border, and the risk is rupture of the free wall, the septum, or a papillary muscle because dead muscle has been removed and collagen has not formed.

· Heart 2 is about 1 to 3 days old: complete coagulative necrosis with brisk neutrophils, and the linked complication is fibrinous pericarditis over a transmural infarct.

· Heart 3 is about 10 to 14 days old: established granulation tissue with new vessels and collagen, and rupture is now less likely while aneurysm and mural thrombus become the concerns as the scar forms.

· Heart 4 is over 2 months old: dense collagen in a firm white scar, and the risks are heart failure and scar-related ventricular tachycardia.

· Heart 1 has the weakest wall. The wall is weakest after macrophages have removed the dead muscle and before collagen has been laid down, about days 3 to 10, which is the window for rupture.

· A pale unstained zone on tetrazolium staining with normal light microscopy means the infarct is younger than the point at which necrosis becomes visible, roughly 2 to 3 hours onward, because loss of dehydrogenase activity comes first.

Distinctions that matter

· Strength of the wall tracks the balance between removal of necrotic muscle and deposition of collagen, which is why the softest stage is also the one that ruptures.

· Each stage depends on the one before, so the appearance of an infarct dates it and the date predicts the complication.

Common misconceptions

· Assuming a bigger or older infarct is more likely to rupture. Rupture belongs to the narrow window when the muscle is gone and the scar is not yet built.

· Dating an infarct by gross appearance alone. The inflammatory and repair cells on microscopy are the more reliable clock.

Where it is taught: Section C05, morphologic timeline table and the closing paragraph on wall strength of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C06-BD1

Brain dump 1 · foundational retrieval: Complications laid out on a time line

A competent response contains

· First minutes to hours: ventricular fibrillation, the main cause of death in the first hours; inferior infarcts also give sinus bradycardia and atrioventricular block.

· First hours to days: pump failure with pulmonary congestion, cardiogenic shock, and right ventricular infarction from proximal right coronary occlusion.

· About 1 to 3 days: fibrinous or fibrinohemorrhagic pericarditis over a transmural infarct, with pleuritic positional pain and a friction rub.

· About 3 to 7 days, from day 1 to about 2 weeks or later: rupture of the free wall, the septum, or a papillary muscle.

· Weeks: mural thrombus over a large anterior apical infarct, Dressler syndrome, and true aneurysm; months onward, heart failure and scar-related ventricular tachycardia.

· The three electrical changes in ischemic myocytes are a less negative resting potential, slow conduction, and altered refractoriness, so a wavefront can block in one direction and re-enter tissue it has already passed, which is the substrate for reentry.

· Added instability comes from abnormal automaticity, triggered activity, local hyperkalemia, and catecholamines.

· Loss of about 40% or more of the left ventricle causes cardiogenic shock.

· An inferior infarct slows the heart because the right coronary artery supplies the nodes and because vagal reflexes arise from the inferior and posterior wall; bundle branch block from an anterior septal infarct is more ominous because it reflects extensive necrosis below the atrioventricular node.

· Cardiogenic shock cycle: loss of about 40% or more of left ventricular muscle; stroke volume and cardiac output fall; aortic pressure falls while left ventricular diastolic pressure rises; coronary perfusion pressure falls; ischemia extends into surviving muscle; contractility falls further.

· Right ventricular infarction gives hypotension, distended neck veins, and clear lungs, because the right ventricle cannot deliver blood to the left heart.

Distinctions that matter

· Early complications are electrical and hemodynamic, middle ones are mechanical and follow tissue weakness, and late ones follow the scar.

· Inferior bradycardia is partly reflex and metabolic and usually recovers, while block from an anterior septal infarct reflects dead conducting tissue.

Common misconceptions

· Treating every post-infarction bradycardia as the same problem. Site decides level, mechanism, and outlook.

· Expecting rupture in the first day or two. The wall is still held together by necrotic muscle until macrophages clear it.

Where it is taught: Section C06, paragraphs on arrhythmia and pump failure, the cardiogenic shock cycle, and the mechanical complications table; the timeline in Section C05 of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C06-BD2

Brain dump 2 · integration and application: Four patients deteriorate after an infarction

A competent response contains

· Patient 1: papillary muscle rupture with acute severe mitral regurgitation, giving sudden pulmonary edema and hypotension with an apical murmur that may be soft because pressures equalize across the valve.

· Patient 2: septal rupture with a left-to-right shunt, giving a new holosystolic murmur at the left lower sternal border and an oxygen step-up in the right ventricle; anterior infarcts tear the apical septum and inferior infarcts the basal septum.

· Patient 3: free wall rupture with blood in the pericardial sac, giving tamponade with hypotension, distended neck veins, and muffled heart sounds, and pulseless electrical activity; it is classically a first infarct without hypertrophy or collaterals.

· Patient 4: fibrinous pericarditis over a transmural infarct at about 1 to 3 days, with pleuritic positional pain and a friction rub.

· All three ruptures occur mostly at about 3 to 7 days because macrophages have removed the dead muscle and collagen has not yet formed, so the wall is at its weakest.

· The posteromedial papillary muscle ruptures more often because it usually depends on the posterior descending artery alone, while the anterolateral muscle receives both left anterior descending and circumflex branches; rupture of a whole trunk is often rapidly fatal, and ischemia without rupture causes milder regurgitation.

· Pseudoaneurysm is a contained free wall rupture whose wall is pericardium and thrombus, with a narrow neck and a high risk of later rupture; a true aneurysm is a thin transmural scar over a large anterior apical

infarct, appearing weeks to months later, with a wide neck, dyskinesis in systole, and rare rupture, and it carries heart failure, thrombus, ventricular tachycardia, and persistent ST elevation.

· Dressler syndrome is pericarditis several weeks after infarction with fever, pleuritis, and effusion, caused by exposure of sequestered myocardial antigens with antibodies and immune complexes inflaming the pericardium; the delay fits an adaptive immune response, which separates it from the direct fibrinous pericarditis of days 1 to 3.

Distinctions that matter

· The site that tears decides the syndrome: free wall gives tamponade, septum gives a shunt, and papillary muscle gives acute mitral regurgitation.

· A loud murmur is not a measure of severity in acute mitral regurgitation; a soft murmur can accompany torrential regurgitation once left atrial and ventricular pressures equalize.

Common misconceptions

· Expecting a loud murmur in every papillary muscle rupture. Rapid pressure equalization can make it soft.

· Grouping pericarditis at day 2 with Dressler syndrome. One is direct inflammation over necrosis and the other is a delayed immune reaction weeks later.

Where it is taught: Section C06, paragraph on pericarditis, rupture, and thrombus, the mechanical complications table, and the Dressler paragraph of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C06-CA01

Clinical application: Sudden breathlessness on the fifth day after an inferior infarction

A competent response contains

· Recognize: rupture of a papillary muscle with acute severe mitral regurgitation; the structure that failed is the posteromedial papillary muscle.

· Explain: by day 5 macrophages have removed the necrotic myocytes and collagen has not yet been laid down, so the tissue is at its weakest between about days 3 and 10, which is the window in which rupture occurs.

· Differentiate: free wall rupture would give tamponade with distended neck veins, muffled heart sounds, and pulseless electrical activity, and the jugular venous pressure here is not raised; septal rupture would give a holosystolic murmur at the left lower sternal border with an oxygen step-up in the right ventricle rather than a soft apical murmur with pulmonary edema; reinfarction would show new ST elevation and a fresh rise of the MB isoenzyme of creatine kinase, and this tracing shows no new ST change.

· Investigate: look for severe mitral regurgitation with a flail or disrupted papillary muscle on imaging, pulmonary edema on the chest radiograph, and a raised left atrial or pulmonary venous pressure; the murmur can be soft because left atrial pressure rises steeply and the pressure difference driving flow across the valve falls, so little turbulence is generated.

· Decide: the posteromedial papillary muscle has failed, because it usually depends on the posterior descending artery alone while the anterolateral muscle receives branches from both the left anterior descending and the circumflex arteries.

· Justify: this was an inferior infarct with ST elevation in II, III, and aVF, and the posterior descending artery arises from the right coronary artery in about 80% of hearts, so the inferior wall and the posteromedial

papillary muscle infarct together; troponin rises again because fresh myocyte necrosis accompanies the rupture, while the absence of new ST elevation shows there is no new coronary occlusion.

Distinctions that matter

· Clear neck veins with flooded lungs point to a lesion that drives blood backward into the left atrium, while raised neck veins with clear lungs point to the right side or to the pericardium.

· Rising troponin without new ST change fits new tissue destruction from a mechanical complication rather than a new occlusion.

Common misconceptions

· Reading sudden pulmonary edema after an infarction as simple pump failure. A new murmur with abrupt onset on day 5 points to a mechanical complication.

· Using murmur loudness to grade the regurgitation. The murmur can be soft when pressures equalize quickly.

Where it is taught: Section C06, paragraph on pericarditis, rupture, and thrombus, the mechanical complications table, and the stop and predict box; Section C04 for the territory and the posterior descending artery of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C06-Q02

Board-style question: Timing of free wall rupture

Correct answer: A. Macrophages have removed the necrotic myocytes and collagen deposition has not yet strengthened the wall

Why it is right: Between about days 3 and 10 the necrotic myocytes have been phagocytosed and granulation tissue has not yet deposited enough collagen to take their place, so the infarcted wall is at its mechanically weakest and can tear under systolic pressure, which is why rupture clusters in that window.

B is wrong: Digestion of a coronary plaque cap by inflammatory enzymes is the mechanism that starts an infarction by causing thrombosis, and it says nothing about the strength of the ventricular wall 5 days later.

C is wrong: A thin fibrous scar that bulges during systole describes a true aneurysm, which forms over weeks to months and rarely ruptures; at 5 days no scar has yet matured.

D is wrong: Slow conduction through dense collagen in a healed infarct sets up reentrant ventricular tachycardia months later and does not weaken the wall or cause a tear.

A competent response contains

· Rupture occurs mostly at about 3 to 7 days, when macrophages have cleared the necrotic myocytes and collagen has not yet formed, so the wall has lost its old strength and gained none.

· A first infarct without hypertrophy or collaterals is the classic setting for free wall rupture, which fills the pericardial sac and causes tamponade with pulseless electrical activity.

Distinctions that matter

· The weakest moment of the wall is defined by the balance between clearance of dead muscle and deposition of collagen, not by the size of the infarct.

Common misconceptions

· Expecting a transmural infarct to be weakest on the first day. Necrotic muscle still holds the wall together until macrophages remove it.

Where it is taught: Section C05, morphologic timeline table and closing paragraph; Section C06, mechanical complications table of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C07-BD1

Brain dump 1 · foundational retrieval: The chronically ischemic heart and its remodeling cycle

A competent response contains

· Chronic ischemic heart disease, also called ischemic cardiomyopathy, is progressive heart failure from ischemic myocardial damage, usually the late result of one or more healed infarcts, sometimes silent, plus ongoing ischemia from diffuse atherosclerosis; less often it develops without a recognized infarct.

· It is one of the most common causes of heart failure with reduced ejection fraction.

· Gross findings: the heart is enlarged and heavy, with a dilated hypertrophied left ventricle, gray-white scars in coronary territories, moderate to severe coronary atherosclerosis, and sometimes patchy endocardial fibrosis and mural thrombi.

· Microscopic findings: hypertrophied surviving myocytes, subendocardial myocyte vacuolization as a sign of chronic sublethal ischemia, and interstitial and replacement fibrosis.

· Regional scars matching coronary territories separate it from other dilated cardiomyopathies, in which fibrosis is diffuse and the coronary arteries are usually normal.

· Remodeling cycle: infarction and ongoing ischemia reduce contracting muscle; stroke volume falls with sympathetic and renin-angiotensin-aldosterone activation; volume retention and eccentric hypertrophy dilate the left ventricle; a larger radius raises wall stress and oxygen demand; subendocardial ischemia, fibrosis, and myocyte loss worsen; contractile function declines further.

· Angiotensin II and aldosterone promote fibrosis.

· In eccentric hypertrophy myocytes add sarcomeres in series, so the chamber lengthens and enlarges and becomes more spherical.

· Scar cannot contract, so surviving muscle carries the whole load, and myocytes continue to die by apoptosis and ischemia.

Distinctions that matter

· Sarcomeres added in series lengthen the cell and dilate the chamber in eccentric hypertrophy, while sarcomeres added in parallel thicken the wall in concentric hypertrophy.

· Regional scar in arterial territories marks an ischemic cause, while diffuse fibrosis with normal arteries marks other dilated cardiomyopathies.

Common misconceptions

· Treating dilation as purely compensatory. It begins as compensation and then raises wall stress and oxygen demand, which perpetuates the injury.

· Expecting a history of a recognized infarct in every patient. Silent infarcts and diffuse ischemia can produce the same heart.

Where it is taught: Section C07, first two paragraphs, the remodeling paragraph, and the remodeling cycle of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C07-BD2

Brain dump 2 · integration and application: Work the cost of dilation and sort recoverable muscle from scar

A competent response contains

· Wall stress in a roughly spherical ventricle is proportional to pressure times radius divided by twice the wall thickness.

· Normal ventricle: 120 times 2.5 divided by 1.0 gives 300 in these units.

· Remodeled ventricle: 120 times 3.5 divided by 0.8 gives 525 in these units.

· The ratio is 525 divided by 300, about 1.75, so the dilated wall bears about 75% more stress at the same pressure and uses more oxygen for each beat.

· Hypertrophy partly offsets the rise in stress by adding thickness, but it also adds muscle that the narrowed arteries cannot supply.

· Hibernating myocardium is alive, chronically underperfused, and reduces contraction to match the reduced flow, and it can recover if flow is restored.

· Stunned myocardium is alive and already reperfused, and it stays dysfunctional for hours to days before recovering on its own.

· Scar is dead collagenous tissue receiving flow it cannot use, and restoring flow cannot help it; scar is permanent.

· Dilation pulls the papillary muscles apart and enlarges the mitral annulus, so the leaflets no longer meet, giving functional or secondary mitral regurgitation with a normal valve, and the regurgitant volume adds a volume load to an already failing ventricle.

· At scar borders, surviving myocyte strands interwoven with collagen conduct slowly, creating fixed reentry circuits for monomorphic ventricular tachycardia that can degenerate into fibrillation.

· Examination findings include a displaced apical impulse, a third heart sound from rapid filling of a dilated ventricle, and an apical murmur of functional mitral regurgitation; the electrocardiogram shows Q waves from old infarcts.

Distinctions that matter

· Hibernating and stunned myocardium differ by flow: hibernating muscle is still underperfused, while stunned muscle has already been reperfused and simply needs time.

· Functional mitral regurgitation is a geometry problem of the ventricle, while primary mitral regurgitation is a problem of the leaflets or chordae.

Common misconceptions

· Reading poor contraction as dead muscle. Hibernating and stunned myocardium are alive and can recover function.

· Expecting a thicker wall to solve the stress problem. Hypertrophy lowers stress per unit of muscle while adding oxygen demand the narrowed arteries cannot meet.

Where it is taught: Section C07, worked example on the law of Laplace and the paragraphs on recoverable myocardium, valves, and rhythm of the CV-18-C Ischemic Heart Disease notes.